

2021 Year 11 Examination

Mathematics Standard

Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	D
3	A
4	D
5	B
6	A
7	A
8	A
9	B
10	C
11	B
12	D
13	B
14	B
15	C

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Section II

Question 16

Criteria	Mark
• Provides correct solution	1

Sample answer

$$\begin{aligned}384402 \times 1000 &= 384\,402\,000 \text{ m} \\ &= 3.84402 \times 10^8 \text{ m}\end{aligned}$$

Question 17

Criteria	Mark
• Provides correct solution	1

Sample answer

$$\begin{aligned}v^2 &= u^2 + 2as \\ 2as &= v^2 - u^2 \\ a &= \frac{v^2 - u^2}{2s}\end{aligned}$$

Question 18

Criteria	Marks
• Provides correct solution	2
• Provides correct holiday loading for four weeks	1

Sample answer

$$\begin{aligned}\text{Weekly pay} &= \$84\,000 \div 52 \\ &= \$1615.38 \dots\end{aligned}$$

$$\begin{aligned}\text{Pay for 4 weeks} &= 1615.38 \dots \times 4 \\ &= \$6461.54 \dots\end{aligned}$$

$$\begin{aligned}\text{Holiday loading for 4 weeks} &= 17.5\% \times 6461.54 \dots \\ &= \$1130.76 \dots\end{aligned}$$

$$\begin{aligned}\text{Holiday pay for 4 weeks} &= 6461.54 \dots + 1130.76 \dots \\ &= \$7592.31\end{aligned}$$

Question 19

Criteria	Marks
• Provides correct solution	2
• Provides a solution where 3 measures (of mean, median, mode, range) are correct	1

Sample answer

5 numbers: $_ _ _ _ _$
Median = 5: $_ _ 5 _ _$
Mode = 8: $_ _ 5, 8, 8$
Range = 7: $1, _ 5, 8, 8$
Mean = 5: $\frac{1+x+5+8+8}{5} = 5$
 $22 + x = 25$
 $x = 3$

The 5 numbers in this data set are: 1, 3, 5, 8, 8

Question 20 (a)

Criteria	Mark
• Provides the correct answer	1

Sample answer

Highest mark in class A = 99

Question 20 (b)

Criteria	Mark
• Provides correct solution	1

Sample answer

$$\begin{aligned}\text{Percentage} &= \frac{3}{25} \times 100 \\ &= 12\%\end{aligned}$$

Question 20 (c)

Criteria	Marks
Provides correct answer and calculates range or standard deviation to justify	2
Calculates range or standard deviation correctly with incorrect or no conclusion	1

Sample answer

Class A:

Range = 72

Standard Deviation = 18.289...

Class B:

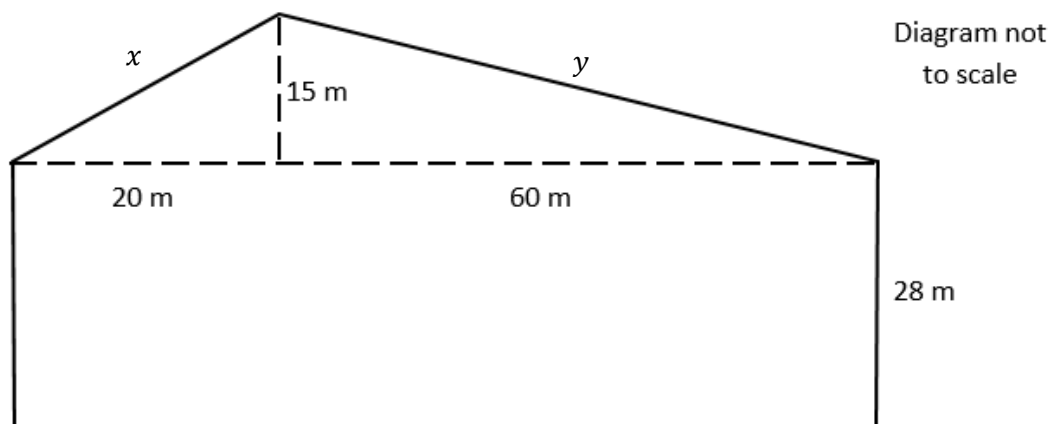
Range = 55

Standard Deviation = 15.398...

Class B has performed more consistently

Question 21 (a)

Criteria	Marks
Provides correct solution	2
Correctly uses Pythagoras' theorem to calculate one unknown length	1

Sample answer

$$x = \sqrt{20^2 + 15^2}$$

$$= 25 \text{ m}$$

$$y = \sqrt{60^2 + 15^2}$$

$$= 61.8 \dots \text{ m}$$

$$\begin{aligned} \text{Perimeter} &= 25 + 61.8 \dots + 28 + 28 + 60 + 20 \\ &= 222.8 \dots \text{ m} \\ &= 223 \text{ m (to the nearest metre)} \end{aligned}$$

Question 21 (b)

Criteria	Marks
• Uses perimeter found in part (a) to correctly determine total cost	2
• Calculates cost per metre, or equivalent merit	1

Sample answer

$$54 + 42 = \$96 \text{ per metre}$$

$$\text{Total cost} = 223 \times 96$$

$$= \$21\,408$$

Question 21 (c)

Criteria	Marks
• Provides correct solution	2
• Correctly calculates upper and lower limits for one length	1

Sample answer

$$\text{Precision} = 1 \text{ m}$$

$$\text{Absolute error} = 0.5 \text{ m}$$

For 80 m length:

$$\text{Lower limit} = 79.5 \text{ m}$$

$$\text{Upper limit} = 80.5 \text{ m}$$

For 28 m length:

$$\text{Lower limit} = 27.5 \text{ m}$$

$$\text{Upper limit} = 28.5 \text{ m}$$

For Area:

$$\begin{aligned} \text{Lower limit} &= 79.5 \times 27.5 \\ &= 2186.25 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Upper limit} &= 80.5 \times 28.5 \\ &= 2294.25 \text{ m}^2 \end{aligned}$$

Question 22 (a)

Criteria	Marks
• Provides correct solution	3
• Provides three or four correct values	2
• Provides one or two correct values	1

Sample answer

6.88 6.93 6.95 6.98 7.07 7.10 7.15 7.30

Lower Extreme = 6.88

Lower Quartile = 6.94

Median = 7.025

Upper Quartile = 7.125

Upper Extreme = 7.30

Question 22 (b)

Criteria	Mark
• Provides the correct answer	1

Sample answer

$\sigma = 0.12913 \dots$

$= 0.129$ (to 3 significant figures)

Question 22 (c)

Criteria	Marks
• Provides correct solution with justification	2
• Correct working with incorrect conclusion	1

Sample answer

$IQR = 7.125 - 6.94$ (from part (a))

$= 0.185$

$$Q_1 - 1.5 \times IQR = 6.94 - 1.5 \times 0.185 \\ = 6.6625$$

$$Q_3 + 1.5 \times IQR = 7.125 + 1.5 \times 0.185 \\ = 7.4025$$

There are no scores outside this range.

$\therefore$ There are no outliers.

Question 22 (d)

Criteria	Marks
• Correctly explains the effect of outliers on the mean and median	2
• Correctly explains the effect of outliers on the mean OR median	1

Sample answer

Outliers will have a large effect on the mean.

Outliers may have a small effect on the median.

Question 23 (a)

Criteria	Mark
• Provides the correct answer	1

Sample answer

14:41 to 19:58

5 hours and 17 minutes.

Question 23 (b)

Criteria	Mark
• Provides the correct answer	1

Sample answer

09:53

Question 23 (c)

Criteria	Mark
• Provides the correct answer	1

Sample answer

04:55

Question 24

Criteria	Marks
• Provides correct solution	2
• Correctly calculates the area of the square cross-section	1

Sample answer

$$\text{Radius} = 10 \text{ cm}$$

$$A_{\text{square}} = 4 \times \left(\frac{1}{2} \times 10 \times 10\right) \\ = 200 \text{ cm}^2$$

$$V = Ah$$

$$2400 = 200 \times h$$

$$h = 2400 \div 200$$

$$h = 12 \text{ cm}$$

Question 25 (a)

Criteria	Marks
• Provides correct solution	2
• Correctly calculates the amount of energy used per year	1

Sample answer

$$\text{Energy usage} = 140 \times 4 \times 365 \\ = 204400 \text{ W/year}$$

$$204400 \div 1000 = 204.4 \text{ kW/year}$$

$$\text{Cost} = 204.4 \times 0.12 \\ = \$24.528 \\ = \$24.53$$

Question 25 (b)

Criteria	Mark
• Provides correct solution	1

Sample answer

$$\frac{50}{140} \times 100 = 35.7\% \text{ (to 1 decimal place)}$$

Question 26 (a)

Criteria	Mark
Provides correct solution	1

Sample answer

$$\begin{aligned}\text{Value after 3 years} &= 22000 - 3 \times 850 \\ &= \$19450\end{aligned}$$

Question 26 (b)

Criteria	Mark
Provides correct solution	1

Sample answer

$$\begin{aligned}\text{Percentage decrease} &= \frac{3 \times 850}{22000} \times 100 \\ &= 11.59\% \text{ (to 2 decimal places)}\end{aligned}$$

Question 26 (c)

Criteria	Marks
Provides correct solution	4
Correctly calculates total amount owing, or equivalent merit	3
Correctly calculates interest, or equivalent merit	2
Correctly calculates loan amount, or equivalent merit	1

Sample answer

$$\begin{aligned}\text{Deposit} &= 10\% \times 22000 \\ &= \$2200\end{aligned}$$

$$\begin{aligned}\text{Loan amount} &= 22000 - 2200 \\ &= \$19800\end{aligned}$$

$$\begin{aligned}\text{Simple Interest} &= 19800 \times 6.5\% \times 5 \\ &= \$6435\end{aligned}$$

$$\begin{aligned}\text{Total amount to repay} &= 19800 + 6435 \\ &= \$26235\end{aligned}$$

$$\begin{aligned}\text{Monthly repayment} &= 26235 \div 60 \\ &= \$437.25\end{aligned}$$

Question 26 (d)

Criteria	Marks
• Provides correct solution	2
• Correctly calculates petrol usage per week, or equivalent merit	1

Sample answer

$$\begin{aligned}\text{Petrol usage per week} &= 7.2 \times 2.5 \\ &= 18 \text{ L}\end{aligned}$$

$$\begin{aligned}\text{Cost per year} &= 18 \times 52 \times 1.42 \\ &= \$1329.12\end{aligned}$$

Question 27

Criteria	Marks
• Provides correct solution	2
• Correctly calculates the probability of drawing a green marble	1

Sample answer

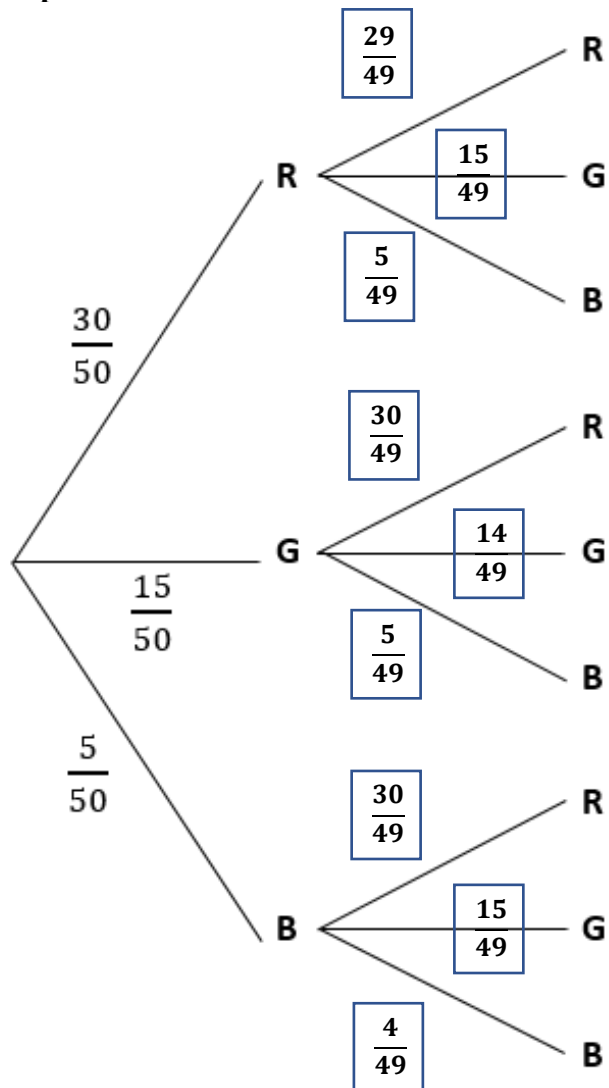
$$\begin{aligned}P(\text{green}) &= 1 - 22\% - \frac{5}{8} \\ &= 0.155\end{aligned}$$

$$\begin{aligned}\text{Exp}(\text{green}) &= 0.155 \times 400 \\ &= 62\end{aligned}$$

Question 28 (a)

Criteria	Marks
Provides correct solution	2
Provides 5 correct probabilities	1

Sample answer



Question 28 (b)

Criteria	Marks
• Correctly uses probabilities in part (a) to determine solution	2
• Correctly calculates the probability of two red OR two green OR two blue, or equivalent merit	1

Sample answer

$$P(\text{same colour}) = P(RR) + P(GG) + P(BB)$$

$$\begin{aligned} &= \frac{30}{50} \times \frac{29}{49} + \frac{15}{50} \times \frac{14}{49} + \frac{5}{50} \times \frac{4}{49} \\ &= \frac{22}{49} \end{aligned}$$

Question 29

Criteria	Marks
Provides correct solution	5
Correctly calculates taxable income, Medicare levy, total tax payable and total tax paid, or equivalent merit	4
Shows correct working for three of the following: taxable income, Medicare levy, total tax payable and total tax paid, or equivalent merit	3
Shows correct working for two of the following: taxable income, Medicare levy, total tax payable and total tax paid, or equivalent merit	2
Shows correct working for one of the following: taxable income, Medicare levy, total tax payable and total tax paid, or equivalent merit	1

Sample answer

$$\begin{aligned}\text{Annual income} &= 2150 \times 26 \\ &= \$55900\end{aligned}$$

$$\begin{aligned}\text{Taxable income} &= 55900 - 550 \\ &= \$55350\end{aligned}$$

$$\begin{aligned}\text{Tax on this income} &= 5092 + 0.325 \times (55350 - 45000) \\ &= \$8455.75\end{aligned}$$

$$\begin{aligned}\text{Medicare levy} &= 2\% \times 55350 \\ &= \$1107\end{aligned}$$

$$\begin{aligned}\text{Total tax payable} &= 8455.75 + 1107 \\ &= \$9562.75\end{aligned}$$

$$\begin{aligned}\text{Tax paid} &= 352 \times 26 \\ &= \$9152\end{aligned}$$

$$\begin{aligned}\text{Amount owing} &= 9562.75 - 9152 \\ &= \$410.75\end{aligned}$$

Andrea will have to pay \$410.75 more tax.

Question 30 (a)

Criteria	Marks
Provides correct solution	2
Correctly applies one application of the trapezoidal rule, or equivalent merit	1

Sample answer

$$5.4 \div 2 = 2.7$$

$$\begin{aligned}\text{Total area} &= \frac{2.7}{2}(4 + 3.2) + \frac{2.7}{2}(3.2 + 3.4) \\ &= 18.63 \text{ m}^2\end{aligned}$$

Question 30 (b)

Criteria	Marks
Correctly uses area in part (a) to calculate volume in litres	2
Correctly calculates volume in cubic metres	1

Sample answer

$$\begin{aligned}\text{Volume} &= 18.63 \times 0.04 \\ &= 0.7452 \text{ m}^3 \\ &= 0.7452 \text{ kL} \\ &= 745.2 \text{ L}\end{aligned}$$

Question 31 (a)

Criteria	Marks
Correct value of k determined with a description of what it represents	2
Correct value of k determined OR a correct description of what k represents	1

Sample answer

When $t = 10$, $W = 2$

$$W = kt$$

$$2 = 10k$$

$$k = 0.2$$

The kitchen tap drips at a rate of 0.2 L per hour.

Question 31 (b)

Criteria	Mark
Provides an equation using the constant of variation from part (a)	1

Sample answer

$$W = 0.2t$$

Question 31 (c)

Criteria	Marks
Provides correct solution using equation in part (b)	2
Correct working with incorrect conclusion, or equivalent merit	1

Sample answer

$$2 \text{ weeks} = 2 \times 7 \times 24$$

$$= 336 \text{ hours}$$

$$W = 0.2t$$

$$= 0.2 \times 336$$

$$= 67.2 \text{ L}$$

$$67.2 > 50$$

$\therefore$ The sink will overflow before the family returns from their holiday.

Question 32

Criteria	Marks
Provides correct solution, stating the earliest time either could drive home	5
Correctly calculates the time taken for Laura's BAC to reach zero and Sally's BAC to reach 0.05	4
Correctly calculates BAC and time to reach zero for both Laura and Sally. Calculates time for a BAC of 0.05 to reach zero	3
Correctly calculates BAC and time to reach zero for both Laura and Sally	2
Correctly calculates BAC for both Laura and Sally	1

Sample answer

Laura	Sally
62 kg 4 drinks 4 hours Driving BAC 0 Stopped drinking at 10 pm BAC at 10 pm $= \frac{10 \times 4 - 7.5 \times 4}{5.5 \times 62}$ $= 0.0293255132 \dots$ Time until BAC reaches zero $= \frac{0.0293255132 \dots}{0.015}$ $= 1.955034213 \dots$ $= 1\text{h } 57\text{ min } 18.12\text{s}$ Laura can drive at 11:58 pm (to the nearest minute)	75 kg 6 drinks 4 hours Driving BAC 0.05 Stopped drinking at 10 pm BAC at 10 pm $= \frac{10 \times 6 - 7.5 \times 4}{5.5 \times 75}$ $= 0.0727272727 \dots$ Time until BAC reaches zero $= \frac{0.0727272727 \dots}{0.015}$ $= 4.84848484848 \dots$ $= 4\text{h } 50\text{ min } 54.55\text{s}$ Time for BAC of 0.05 to reach zero $= \frac{0.05}{0.015}$ $= 3\text{h } 20\text{ min}$ Time for BAC of 0.07272... to reach 0.05 $= 4\text{h } 51\text{ min} - 3\text{h } 20\text{ min}$ $= 1\text{h } 31\text{ min (to the nearest minute)}$ Sally can drive at 11:31 pm (to the nearest minute)

Sally can drive home earliest, at 11:31 pm.

2021 Year 11 Mathematics Standard Mapping Grid

Section I

Question	Marks	Content	Syllabus Outcomes
1	1	Significant figures	MS-M1
2	1	Solving an equation	MS-A1
3	1	Percentage error	MS-M1
4	1	Simple interest	MS-F1
5	1	Medication dosage	MS-A1
6	1	Equation of a line	MS-A2
7	1	Box plot	MS-S1
8	1	Elapsed time	MS-M2
9	1	Overtime	MS-F1
10	1	Calculating probability	MS-S2
11	1	Time zones	MS-M2
12	1	Direct variation	MS-A2
13	1	Pareto chart	MS-S1
14	1	GST	MS-F1
15	1	Latitude and longitude	MS-M2

Section II

Question	Marks	Content	Syllabus Outcomes
16	1	Measurement conversion and standard form	MS-M1
17	1	Changing the subject of a formula	MS-A1
18	2	Annual leave loading	MS-F1
19	2	Mode, median, mean, range	MS-S1
20a	1	Reading data display: stem and leaf plot	MS-S1
20b	1	Analysing data display: stem and leaf plot	MS-S1
20c	2	Analysing data display: stem and leaf plot	MS-S1
21a	2	Perimeter, Pythagoras' theorem	MS-M1
21b	2	Application of perimeter	MS-M1
21c	2	Limits of accuracy	MS-M1
22a	3	Five number summary	MS-S1
22b	1	Standard deviation	MS-S1
22c	2	Outliers	MS-S1
22d	2	Effect of outliers on mean and median	MS-S1
23a	1	Timetable	MS-M2
23b	1	Timetable	MS-M2
23c	1	Timetable	MS-M2
24	2	Volume	MS-M1
25a	2	Electricity consumption	MS-M1
25b	1	Electricity consumption	MS-M1
26a	1	Straight line depreciation	MS-F1
26b	1	Percentage decrease	MS-F1
26c	4	Simple interest	MS-F1
26d	2	Fuel consumption	MS-F1
27	2	Expected frequency	MS-S2
28a	2	Probability tree diagram	MS-S2
28b	2	Calculating probabilities	MS-S2
29	5	Tax table	MS-F1
30a	2	Trapezoidal rule	MS-M1
30b	2	Volume and capacity	MS-M1
31a	2	Constant of variation	MS-A2
31b	1	Direct variation	MS-A2
31c	2	Direct variation problem	MS-A2
32	5	Calculate and interpret BAC	MS-A1